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Gate Field-Induced Dynamic Schottky Barrier Height Reduction in Bilayer MoS₂ for Sub-60 mV/dec Schottky Barrier FETs

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ABSTRACT

Bilayer MoS₂ exhibits bandgap narrowing under a vertical electric field due to inversion symmetry breaking, with the extent of reduction scaling proportionally with field strength. Leveraging this intrinsic property, this study investigates its impact on the performance of bilayer MoS₂ Schottky barrier field-effect transistors, with a particular focus on the role of Schottky barrier height reduction in improving subthreshold swing. Density functional theory calculations quantify field-dependent shifts in the conduction and valence band edges, which are integrated into transport simulations considering thermionic emission and tunneling at the metal-semiconductor interface, as well as drift-diffusion in the channel. The barrier height reduction achieves a subthreshold swing of 44.7 mV/dec in a bilayer MoS₂ FET with a 2 nm HfO₂ gate dielectric, representing a 37.5% improvement. In a CMOS inverter configuration, barrier height reduction leads to improvements in switching speed by up to 38% and reduces total power consumption by approximately 5%, demonstrating its effectiveness in enhancing both performance and energy efficiency.

1 | Introduction

2D semiconductors are extensively investigated as potential replacements for traditional 3D counterparts, as they face critical challenges in further scaling and improvements. The main advantages reside in their atomically thin structures. For instance, in field-effect transistors (FETs), it provides excellent electrostatic control and reduced short-channel effects. Among those 2D semiconductors, monolayer molybdenum disulfide (MoS₂), one of the transition metal dichalcogenides (TMDs), has attracted particular interest for logic device applications, owing to its proper intrinsic bandgap (~1.8 eV) [1, 2] and the absence of dangling bonds that minimizes interface traps [3]. Recent advances have demonstrated the feasibility of large-scale 3D integration of monolayer MoS₂ FETs, underscoring its potential for next-generation electronics by enabling high-density circuit

architectures and improved device performance [4]. Yet, any FET structure has the well-known subthreshold swing (SS) limit of 60 mV/dec at room temperature, and 2D semiconductor FETs are not an exception. This limit is dictated by Boltzmann statistics, which may, if it is overcome, bring high-speed and ultra-low-power logic devices [3–5].

Overcoming the subthreshold SS limitation requires innovative approaches that leverage the intrinsic properties of advanced materials. In particular, the unique transport mechanisms at the metal-2D semiconductor interface, along with the distinct electronic characteristics of 2D materials, offer promising pathways toward achieving sub-thermionic SS values. Among these, the layer-dependent electronic properties of TMDs stand out as strong candidates. In bilayer MoS₂, the bandgap transitions from a direct to an indirect type due to interlayer coupling,

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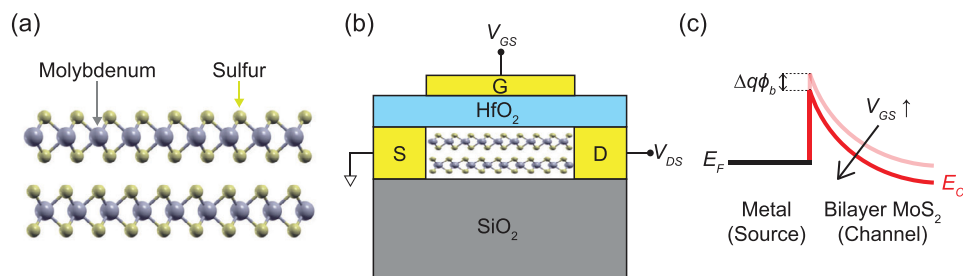


FIGURE 1 | (a) Structure of hexagonal MoS₂ (2H-MoS₂). (b) Architecture of edge-contact bilayer MoS₂ SB-FET. (c) Description of the reduction of the Schottky barrier height near the source contact through the vertical gate field.

and a field-dependent bandgap tunability emerges under an externally applied electric field. This shift is primarily attributed to the breaking of inversion symmetry, which plays a crucial role in determining the electronic band structure. When a vertical electric field is applied, this symmetry is broken, leading to a reduction in the bandgap, with the degree of reduction being proportional to the field strength [6–9]. Notably, these bandgap changes can be naturally induced in a FET structure with an edge-contact configuration, which in turn significantly impacts the device's electrostatic control and performance.

This study investigates the potential for improving the SS in bilayer MoS₂ FETs by exploiting the reduction of the Schottky barrier height ($q\phi_b$) in an edge-contact configuration (Figure 1). As shown in Figure 1c, the reduction of $q\phi_b$ according to the vertical field by gate bias can dynamically enhance the current injection. Density functional theory (DFT) simulations confirm that the bandgap of bilayer MoS₂ narrows under an applied vertical electric field, consistent with previously reported field-induced symmetry-breaking effects. Building upon this, a transport model that incorporates both the metal-2D semiconductor interface physics and the field-dependent band structure of bilayer MoS₂ is developed to analyze carrier dynamics. Unlike prior studies that rely on ballistic transport models [10–12], which are primarily applicable to FETs with gate lengths of just a few nanometers, this model captures drift-diffusion transport behavior, making it more relevant for realistic device dimensions. Simulation results demonstrate that the gate-induced reduction in $q\phi_b$ leads to a substantial increase in subthreshold current. This effect enables SS values below the thermionic limit of 60 mV/dec, highlighting the feasibility of bilayer MoS₂ Schottky barrier FETs (SB-FETs) with edge-contact configurations for ultra-low-power logic applications.

2 | Results and Discussion

2.1 | Bandgap Narrowing of Bilayer MoS₂

The bandgap reduction of bilayer MoS₂ mainly stems from the breaking of the inversion symmetry by the external electric field [6–9]. As shown in Figure 2, due to the symmetry breaking, the lowering of the conduction band minimum (CBM) and the raising of the valence band maximum (VBM) occur, and they both contribute to this band narrowing. Figure 2c demonstrates that bandgap reduction scales with vertical electric field strength, yielding an extracted reduction rate of 0.068 eV per V/nm (0.035

from CBM lowering and 0.033 from VBM rising). This value aligns with the 0.066 eV per V/nm reported in a previous study [7].

2.2 | Current–Voltage Characteristics of Schottky Barrier FETs

Unlike conventional metal-oxide-semiconductor FETs (MOS-FETs), which exhibit unipolar transistor characteristics, SB-FETs display ambipolar transistor behavior [13–15]. This characteristic arises from the modulation of carrier injection at the Schottky contacts: a positive gate bias enhances electron tunneling, thereby increasing the drain current, while a negative gate bias facilitates hole tunneling, similarly increasing the current. As shown in Figure 3a, developed transport simulations reproduce this ambipolar transistor behavior. Furthermore, as demonstrated in Figure 3c,d, the simulations accurately capture the dependence of the minimum drain current ($I_{DS,min}$) and the corresponding gate voltage ($V_{GS,min}$) on the applied drain bias (V_{DS}) [13–15]. Under positive V_{DS} , increasing V_{DS} leads to a rise in $I_{DS,min}$ and a positive shift in $V_{GS,min}$. Conversely, under negative V_{DS} , increasing the magnitude of the bias results in a higher $I_{DS,min}$ and a corresponding negative shift in $V_{GS,min}$. Furthermore, comparison with ref. [15] shows good agreement between experimental measurements and transport simulations (Figure 3e).

2.3 | Boltzmann Limit

To investigate the influence of the $q\phi_b$ on bilayer MoS₂ SB-FETs, copper (Cu), a metal with a work function of 4.9 eV, was selected as the source and drain contact material. Along with the energy bandgap (1.4 eV) and the electron affinity (~ 4.2 eV) [16, 17] of bilayer MoS₂, this configuration yields equal Schottky barrier heights for electrons ($q\phi_{b,n}$) and holes ($q\phi_{b,p}$) as 0.7 eV. In addition, the channel length was set to 2 μm . To further explore the relationship between gate controllability and electron transport at the metal-MoS₂ interface, four distinct gate oxide thicknesses (t_{ox}) were employed: 2, 3, 6, and 9 nm.

Without considering bandgap narrowing in bilayer MoS₂, all simulated SB-FETs exhibited SS exceeding the Boltzmann limit of 60 mV/dec (Figure 4). For example, at a drain voltage of $V_{DS} = 10$ mV, the SS values for devices with gate oxide thicknesses of 2, 3, 6, and 9 nm were 71.5, 77.8, 91.3, and 103 mV/dec, respectively. Similar trends were observed at $V_{DS} = -10$ mV, with SS values of 68.7, 73.5, 85.5, and 95.7 mV/dec. This SS degradation appeared

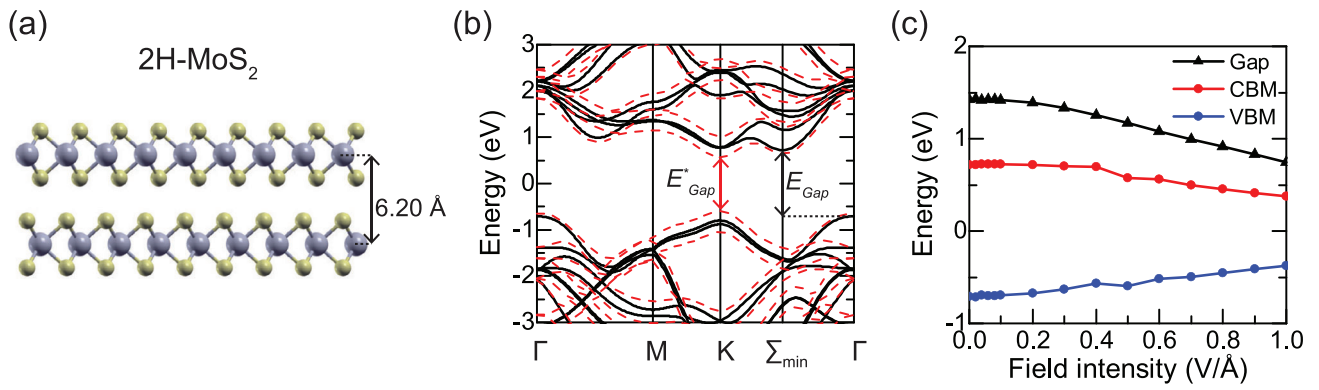


FIGURE 2 | (a) Atomic structure of 2H-MoS₂. The interlayer distances are the results of structural relaxation in the equilibrium state. (b) Energy dispersion diagrams in equilibrium state (black line) and under a vertical field of 0.5 V/Å (red dashed line). (c) Variation in energy bandgap, CBM, and VBM with vertical electric field intensity.

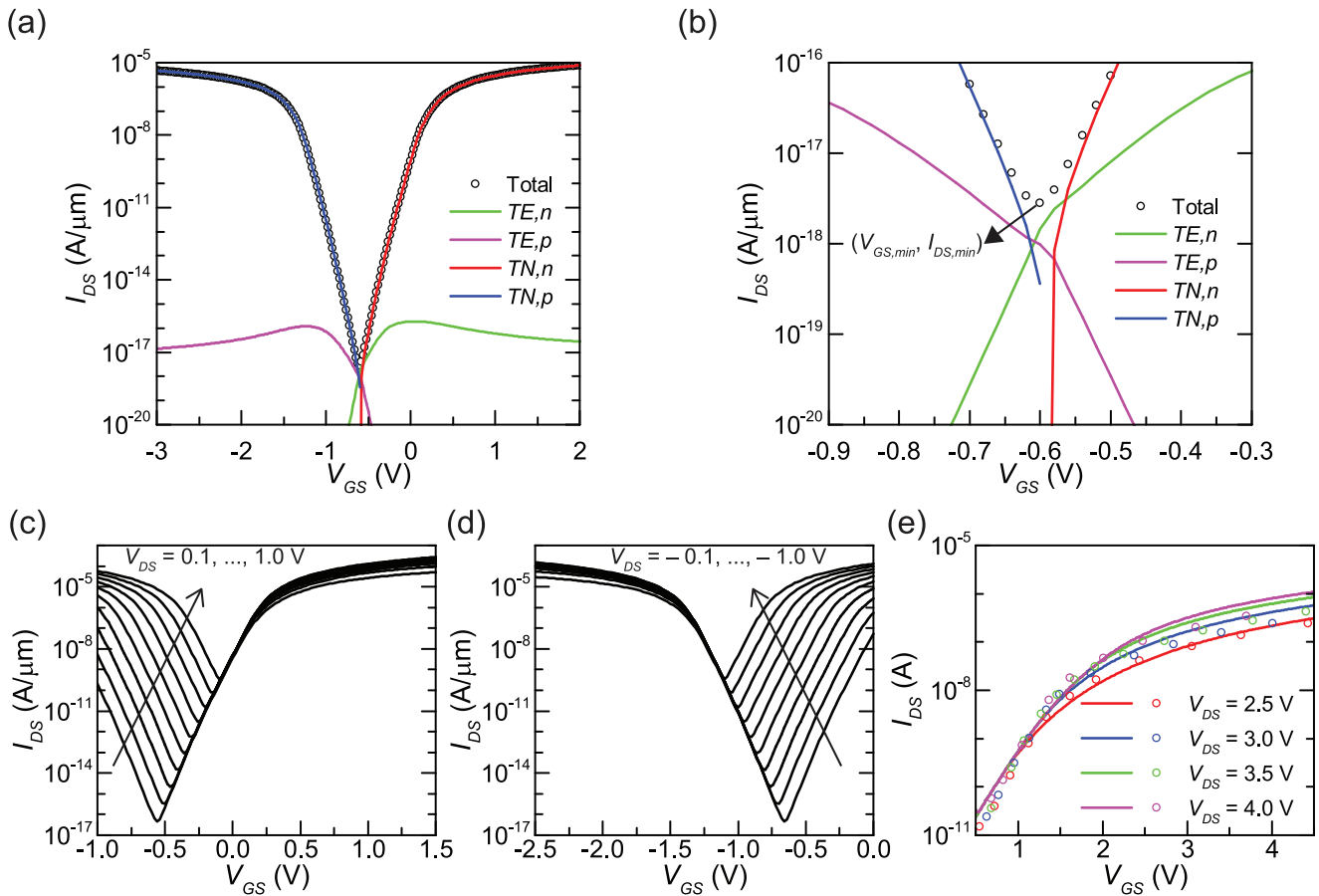


FIGURE 3 | (a) $\log(I_{\text{DS}})$ - V_{GS} characteristic for bilayer MoS₂ SB-FET with a 2 nm HfO₂ gate dielectric at $V_{\text{DS}} = 10$ mV. (b) Magnified view of the $\log(I_{\text{DS}})$ - V_{GS} curve shown in subplot (a). In subplots (a) and (b), the contributions from thermionic emission (TE) and tunneling (TN) mechanisms are indicated. Electron and hole transports are denoted by n and p , respectively. (c) $\log(I_{\text{DS}})$ - V_{GS} characteristic for bilayer MoS₂ SB-FETs with a 2 nm HfO₂ gate dielectric, simulated at positive drain voltages ranging from 0.1 V to 1.0 V in 0.1 V increments. (d) $\log(I_{\text{DS}})$ - V_{GS} characteristic for bilayer MoS₂ SB-FETs with a 2 nm HfO₂ gate dielectric, simulated at negative drain voltages from -0.1 V to -1.0 V in -0.1 V increments. (e) Transfer characteristics of tri-layer MoTe₂ SB-FETs. Experimental data from ref. [15] are shown as symbols. Solid lines represent results obtained from transport simulations.

in both thermionic emission and tunneling regimes. However, tunneling posed a greater challenge, as revealed by decomposing the total current into its constituent components. Near the flat-band voltage ($V_{\text{FB}} = -0.6$ V), thermionic emission dominates the injected current. As the gate voltage increases, tunneling current

progressively becomes the dominant transport mechanism. For instance, with a 2 nm HfO₂ gate dielectric (red line in Figure 4a), the tunneling current fraction at the source contact increases from 0% at $V_{\text{gs}} = -0.6$ V to 88.9% at $V_{\text{gs}} = -0.5$ V, reaching 99.0% at $V_{\text{gs}} = -0.4$ V. The subthreshold region spans approximately 0.7 V,

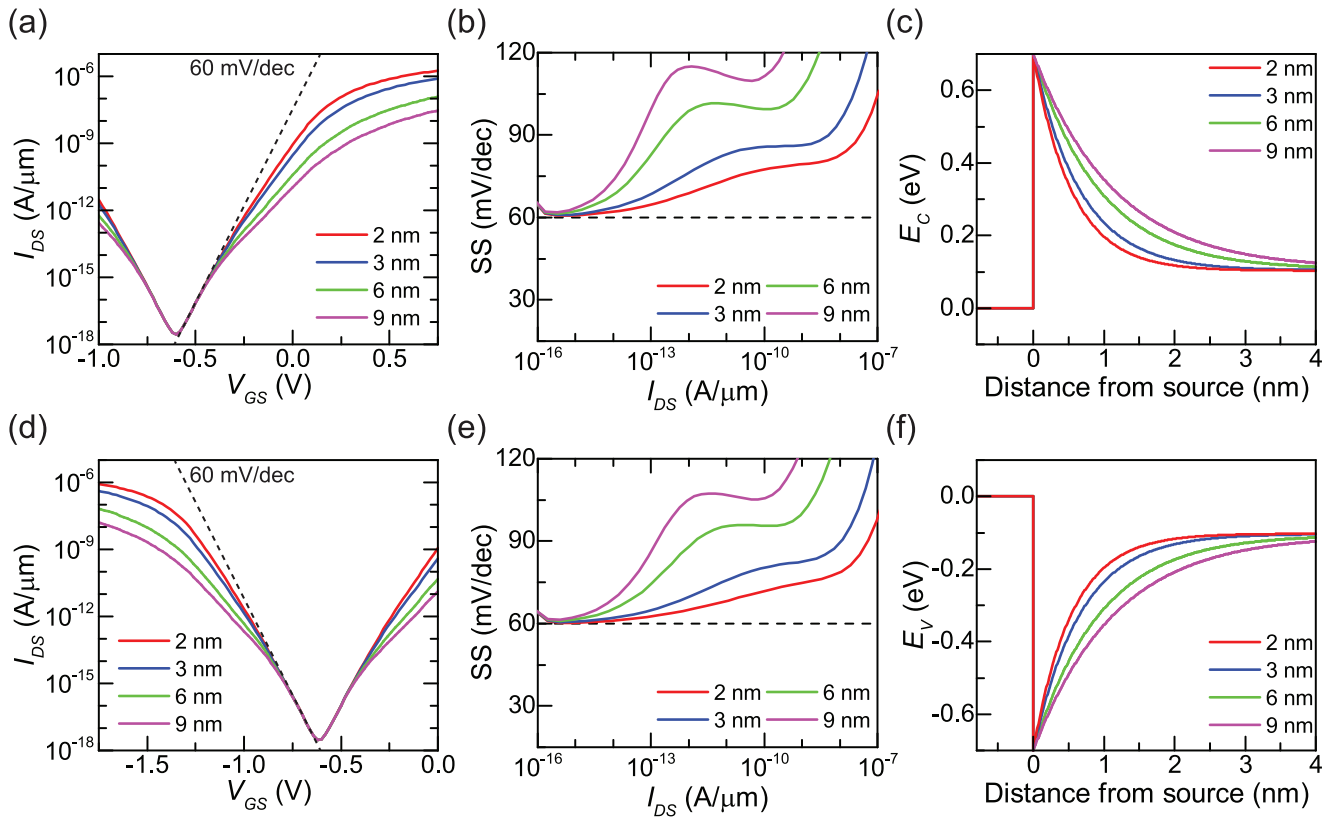


FIGURE 4 | (a) $\log(I_{DS})$ - V_{GS} characteristics of bilayer MoS₂ Schottky-barrier FETs for different oxide thicknesses at $V_{DS} = 10$ mV. (b) Corresponding SS- $\log(I_{DS})$ plots derived from (a). (c) Conduction-band profile near the source contact at $V_{GS} = 0$ V. (d) $\log(I_{DS})$ - V_{GS} characteristics for different oxide thicknesses at $V_{DS} = -10$ mV. (e) Corresponding SS- $\log(I_{DS})$ plots derived from (d). (f) Valence-band profile near the source contact at $V_{GS} = -1.2$ V. All results in this figure were obtained without including bandgap narrowing effects in bilayer MoS₂.

from -0.6 to 0.1 V, within which the average subthreshold swing increases due to the growing influence of tunneling current, which impacts the slope of the current-voltage characteristics. In the thermionic emission-dominated regime, the SS approaches the Boltzmann limit due to the energy distribution of injected carriers following the classical Maxwell-Boltzmann statistics, largely unaffected by barrier shape or thickness (Figure 4a,d). In contrast, as tunneling becomes significant, the SS begins to correlate strongly with gate controllability. Specifically, reducing the tunneling barrier width—achievable via thinner oxides or more aggressive electrostatic gating—can steepen the SS substantially (Figure 4c,f). Nevertheless, the SS in the tunneling regime without bandgap narrowing remains worse than in the thermionic regime. This reflects a fundamental difference in transport physics: thermionic emission depends primarily on carrier energy, whereas tunneling is governed by barrier width and height. Thus, improving tunneling-controlled subthreshold swing requires precise electrostatic engineering of the barrier shape, not just thermal modulation.

2.4 | Subthreshold Swing of Sub-60 mV/dec

A substantial improvement of SS was observed when the reduction of the barrier height was included, which was induced by the vertical electric field near the metal-semiconductor junction. To quantitatively describe this effect, the reduction rate of the Schottky barrier height was characterized as $\beta_{\phi,u} = \Delta\phi_{b,u}/\Delta V_{GS}$

[V/V], where ΔV_{GS} represents the gate bias relative to the flat-band voltage and $\Delta\phi_{b,u}$ represents the Schottky barrier height relative to the flat-band condition. In the barrier modulation factor, $\beta_{\phi,u}$, u can be n or p , which are related to the CBM lowering and VBM rising. As depicted in Figure 5, a larger β_{ϕ} , indicative of an increased rate of barrier height reduction in response to the gate bias, leads to a more pronounced steepness in the $\log(I_{DS})$ - V_{GS} characteristic in the sub-threshold regime. Since the electric field intensity experienced by the channel in a FET structure depends on the relative permittivity of the gate dielectric, the bandgap narrowing rate extracted from DFT calculations was scaled by a factor of 25, corresponding to the relative permittivity of HfO₂. Consequently, the CBM lowering and VBM rising ratios become 0.875 and 0.825 V per V/nm, respectively. Accordingly, the values of β_{ϕ} for electrons and holes are calculated using the expression $\beta_{\phi,n} = 0.875/t_{ox}$ and $\beta_{\phi,p} = 0.825/t_{ox}$, respectively. The corresponding values for various t_{ox} are summarized in Table 1. Considering the flat-band voltage, -0.6 V, the reduction rate of the barrier height was effectively captured through the parameter β_{ϕ} . Additionally, the effective intrinsic carrier density (n_{ie}) due to bandgap narrowing was incorporated using the expression $n_i \cdot \exp(\Delta E_{gap}/2k_B T)$, where n_i represents the intrinsic carrier density.

Further examination of the SS improvement reveals that the SS values at different gate oxide thicknesses at $V_{DS} = 10$ mV were 44.7 , 53.3 , 72.8 , and 90.0 mV/dec for 2 , 3 , 6 , and 9 nm, respectively. In addition, the SS for different gate oxide thicknesses at $V_{DS} = -10$ mV were extracted as 44.4 , 51.9 , 69.3 , and 85.2 mV/dec for 2 ,

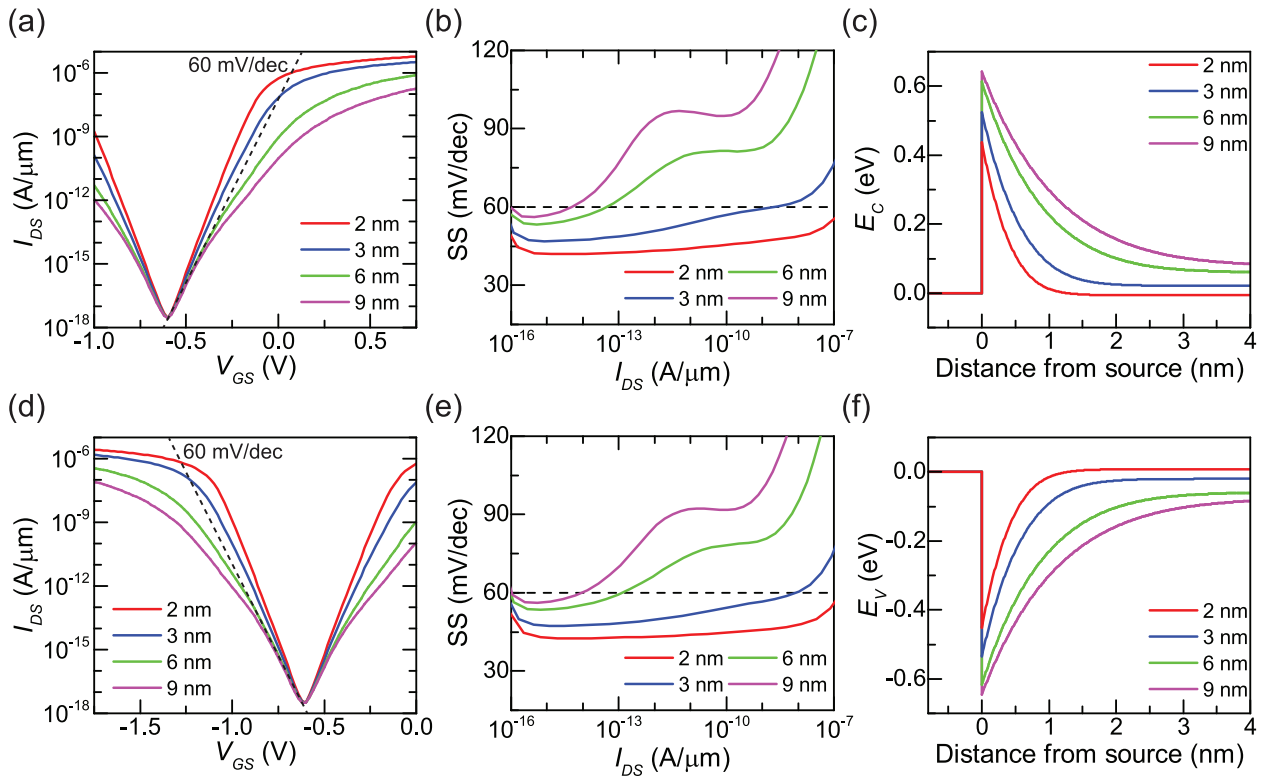


FIGURE 5 | (a) $\log(I_{DS})$ - V_{GS} characteristic for each oxide thickness of bilayer MoS₂ SB-FETs at $V_{DS} = 10$ mV with application of bandgap narrowing and barrier height reduction. (b) SS- $\log(I_{DS})$ plot derived from the data in (a). (c) Conduction band profile near the source contact at $V_{GS} = 0$ V. (d) $\log(I_{DS})$ - V_{GS} characteristic for each oxide thickness of bilayer MoS₂ SB-FETs at $V_{DS} = -10$ mV with bandgap narrowing and barrier height reduction. (e) SS- $\log(I_{DS})$ plot derived from the data in (d). (f) Valence band profile near the source contact at $V_{GS} = -1.2$ V.

TABLE 1 | Comparison of SS and reduction rate for each oxide thickness.

t_{ox}	$V_{DS} = 10$ mV				$V_{DS} = -10$ mV			
	β_{ϕ_n} [V/V]	SS without ϕ_b reduction	SS with ϕ_b reduction	SS reduction rate	β_{ϕ_p} [V/V]	SS without ϕ_b reduction	SS with ϕ_b reduction	SS reduction rate
		[mV/dec]	[mV/dec]			[mV/dec]	[mV/dec]	
2 nm	0.438	71.5	44.7	37.5%	0.413	68.7	44.4	35.4%
3 nm	0.292	77.8	53.3	31.5%	0.275	73.5	51.9	29.4%
6 nm	0.146	91.2	72.8	20.2%	0.138	85.5	69.3	18.9%
9 nm	0.097	103	90.0	12.6%	0.092	95.7	85.2	11.0%

3, 6, and 9 nm, respectively. A comparison of these values with those obtained prior to barrier height reduction is summarized in Table 1.

The improvement of SS appeared in both the thermionic emission and tunneling regimes. To systematically evaluate the impact of barrier height reduction on SS, SS in both regimes, denoted as SS_{TE} and SS_{TN} , were analysed separately. SS_{TE} was impacted by only β_{ϕ} . In an ideal scenario, the sub-threshold current induced by thermionic emission can be described by the equations:

$$I_{TE} = I_0 \cdot \exp(-q\phi_b/k_B T) \exp(qV_{GS}/k_B T)$$

Accordingly, the SS_{TE} can be found as $dV_{GS}/d\log(I) = 10^3 \cdot k_B T/q \cdot \ln 10 = 60$ [mV/dec]. As such, without barrier height reduction,

the SS_{TE} of any FET cannot go beyond 60 [mV/dec]. Here, the term I_0 , known as the scaling factor, together with $\exp(-q\phi_b/k_B T)$, represents the off-current. When the barrier height reduction in the sub-threshold regime ($\phi_b \rightarrow \phi_b - \beta_{\phi} V_{GS}$) is included, SS_{TE} is adjusted to $SS_{TE} = 60/(1 + \beta_{\phi})$ [mV/dec], indicating SS_{TE} can break the Boltzmann limit with the barrier height reduction. On the other hand, the sub-threshold current by tunneling can be described by the equations [18]:

$$I_{TN} = I_0 \cdot f(\phi_b) \exp(-q\phi_b/E_0) \exp(qV_{GS}/E_0)$$

Subsequently, SS_{TN} is found as $10^3 \cdot E_0/q \cdot \ln 10$ [mV/dec], where the value of E_0 [eV] is much greater than $k_B T$. In this case, $f(\phi_b)$ acts as a scaling function that modifies the current based on the tunneling characteristics, and the term $I_0 \cdot f(\phi_b)$

TABLE 2 | Comparison of SS and reduction rate for each pinning factor with a 2 nm HfO₂ gate dielectric.

σ	$V_{DS} = 10 \text{ mV}$			
	$\beta_{\phi,n} [\text{V/V}]$	SS without ϕ_b reduction [mV/dec]	SS with ϕ_b reduction [mV/dec]	SS reduction rate
1 (Schottky-Mott limit)	0.438	71.5	44.7	37.5%
0.75		73.0	45.6	37.5%
0.50		74.9	46.1	38.5%
0.25		76.3	47.1	38.3%
0 (Bardeen limit)		77.0	47.2	38.7%

$\exp(-q\phi_b/E_0)$ represents the off-current. These equations additionally indicate that, in comparison to the thermionic emission process, the tunneling mechanism confronts more substantial hurdles in surpassing the Boltzmann limit, as discussed in the analysis of Figure 4. Converting E_0 to $\eta k_B T$ ($\eta > 1$) allows SS_{TN} to be represented as $\eta \cdot 60$ [mV/dec]. To apply the barrier height reduction to SS_{TN} , $f(\phi_b) \exp(-q\phi_b/E_0)$ can be simplified into $\exp(-q\phi_b/\gamma k_B T)$. Then, SS_{TN} can be expressed as $(1/\eta + \beta_\phi/\gamma)^{-1} \cdot 60$ [mV/dec]. This equation indicates that a thinner gate oxide thickness reduces η due to enhanced gate controllability, whilst increasing β_ϕ simultaneously through a stronger electric field, thereby improving SS_{TN} .

The Schottky barrier height (SBH) is a critical factor affecting device performance in the Cu/MoS₂ system. Based on the observation that the charge neutrality level with respect to the vacuum level (E_{CNL}) at the metal-MoS₂ interface lies below the Fermi level (E_F) [19], the E_{CNL} is assumed to be 5.0 eV ($E_{CNL} - E_F \approx -0.1$ eV). To analyze the Fermi-level pinning behaviour, the Bardeen model was employed [20], which can be expressed as:

$$q\phi_{b,n} = \sigma \cdot (\Phi_{Cu} - E_{CNL}/q) + (E_{CNL}/q - \chi_{MoS_2})$$

where Φ_{Cu} and χ_{MoS_2} correspond to the work function of copper and the electron affinity of bilayer MoS₂, respectively. The validity of applying the Bardeen model to the Cu/MoS₂ interface is supported by ab initio studies [19], which show that interface-induced mid-gap states pin the Fermi level near the charge neutrality level. Consequently, the Schottky barrier height becomes largely insensitive to variations in the metal work function. This strong pinning behavior, characterized by a pinning factor σ as low as 0.1, has also been confirmed experimentally [21, 22]. The good agreement between theoretical and experimental results justifies using the Bardeen model and associated parameters for evaluating the Schottky barrier height and its effect on the SS in Cu/MoS₂ edge-contact transistors. For the pinning factor (σ) ranging from 0 (strong pinning) to 1 (no pinning), the influence of Schottky barrier height reduction on SS was analyzed. Figure 6 and Table 2 collectively indicate that stronger pinning (smaller σ) yields larger SS values, and this trend persists even after implementing barrier height reduction. Notably, within this pinning range, the change in the extent of SS reduction is minimal. Therefore, controlling interface states at the edge contact to minimize Fermi-level pinning is crucial for optimizing device performance.

Notably, the mechanism driving the improvement in SS is an intrinsic feature unique to bilayer 2D materials. Historically, surpassing the SS limit, constrained by fundamental physical laws, has posed significant challenges for conventional materials, with various strategies encountering inherent limitations. 2D tunneling field-effect transistors (TFETs) [23–27] exploit heterostructure-induced vertical band-to-band tunneling (BTBT) mechanisms to achieve SS values below the thermionic limit in localized regions of the subthreshold regime; however, this enhanced performance is not maintained over the entire subthreshold range. For instance, a previously reported SnSe₂/WSe₂ heterostructure demonstrated a minimum SS of 29 mV/dec, although the sub-60 mV/dec SS was sustained only across a drain current variation of about two orders of magnitude [27]. In contrast, the bilayer MoS₂ SB-FET with a 2 nm HfO₂ dielectric (Figure 5b) achieves sub-60 mV/dec SS over a drain current variation exceeding nine orders of magnitude. Additionally, the BTBT mechanism inherently limits the on-current, posing significant challenges for the practical application of TFETs in low-power electronics. In contrast, the bilayer MoS₂ SB-FET exhibits high on-current through tunneling at the Schottky contact. An alternative strategy employs external functional components, such as ferroelectric materials [28–34] or threshold-switch [35–39]. Ferroelectric materials utilize negative capacitance (NC) effects, while threshold-switch devices exploit negative differential resistance (NDR) effects to amplify the channel potential. For instance, a study employing Al-doped HfO₂ as a ferroelectric layer to induce negative capacitance in monolayer MoS₂ FET achieved a SS of 57 mV/dec; however, the sub-60 mV/dec SS was maintained only across a drain current variation of about two orders of magnitude [30]. While these approaches can indeed achieve SS values below 60 mV/dec, they face significant drawbacks related to increased process complexity, compromised stability, and limited scalability. Specifically, these methods require additional complex fabrication steps to integrate external ferroelectric materials or switching components, adding manufacturing challenges and cost. From a stability perspective, ferroelectric materials commonly suffer from reliability and durability issues, while threshold switches experience performance degradation due to material fatigue and associated hysteresis effects. Furthermore, scalability is constrained since these external components—especially threshold switches connected in series with source or drain terminals—hinder device miniaturization and prevent high-density integration. In contrast, the electric-field-induced bandgap narrowing method leverages intrinsic

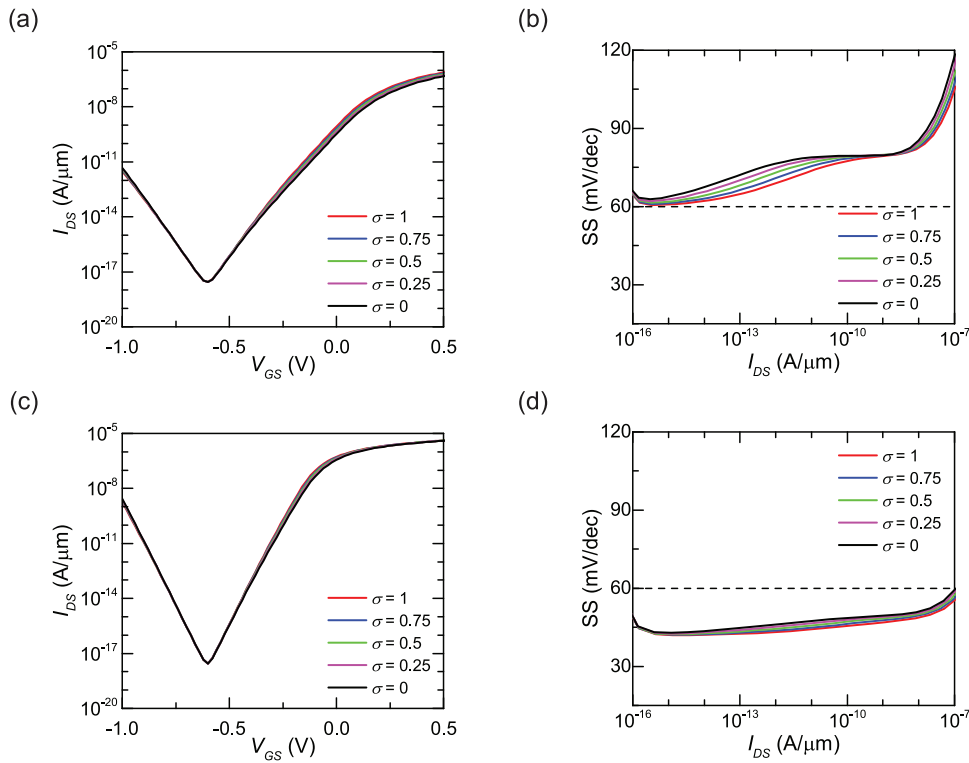


FIGURE 6 | (a) $\log(I_{DS})$ - V_{GS} characteristic of bilayer MoS_2 SB-FETs with a 2 nm HfO_2 gate dielectric at $V_{DS} = 10$ mV for each pinning factor σ . (b) SS- $\log(I_{DS})$ plot derived from the data in (a). (c) $\log(I_{DS})$ - V_{GS} characteristic of bilayer MoS_2 SB-FETs with a 2 nm HfO_2 gate dielectric at $V_{DS} = 10$ mV for each pinning factor σ , considering bandgap narrowing and barrier height reduction. (d) SS- $\log(I_{DS})$ plot derived from the data in (d).

properties of bilayer TMDs, resulting in a simpler fabrication process, enhanced device robustness, and superior scalability. Furthermore, it employs a general edge-contact configuration, naturally formed during metal contact fabrication, providing a distinct advantage in fabrication efficiency.

2.5 | Improved CMOS Inverter Characteristics

The application of Schottky barrier height reduction significantly enhances both the static and dynamic performance of bilayer MoS_2 -based complementary MOS (CMOS) inverters (Figure 7a). In terms of static characteristics (Figure 7b), the inverter initially exhibits a transition range of 0.108 V (from 0.9 V_{DD} to 0.1 V_{DD}), corresponding to 21.6% of V_{DD} . After applying Schottky barrier height reduction, this range is sharply reduced to 0.042 V (8.4% of V_{DD}), indicating a substantially steeper and more defined switching behavior. The dynamic response (Figure 7c,d) also shows pronounced improvements: prior to the application of these effects, the rise delay time (0 to 0.9 V_{DD}) is 4.63 ns, and the fall delay time (V_{DD} to 0.1 V_{DD}) is 4.24 ns. Following the barrier height reduction, the rise and fall delay times are reduced to 3.07 and 2.63 ns, respectively, representing reductions of 33.69% and 37.97%. In terms of power consumption, assuming a 2 nm HfO_2 gate dielectric and a channel width of 100 nm, the dynamic power consumption remains unaffected by the barrier height reduction, as it is defined by the expression [40]:

$$P_{\text{dynamic}} = \alpha C_L V_{DD}^2 f$$

where α is active factor, C_L is the load capacitance, and f is clock frequency. For $\alpha = 0.1$, the dynamic power consumption remains constant at 5.21 nW. In contrast, the static power consumption, given by $P_{\text{static}} = V_{DD} I_{\text{off}}$ [40], is significantly reduced from 272 pW in the baseline CMOS inverter to 0.086 pW with barrier height reduction, corresponding to a reduction of approximately 99.97%. When both dynamic and static components are considered, the total power consumption decreases from 5.48 nW in the baseline configuration to 5.21 nW with barrier height reduction, representing an overall reduction of 4.93%. These results clearly demonstrate that bandgap narrowing and Schottky barrier height engineering improve both switching speed and energy efficiency in bilayer MoS_2 -based CMOS inverters.

The proposed Schottky barrier height engineering approach is expected to benefit beyond CMOS inverters, extending to more complex logic gates such as NAND and XOR, as well as memory devices like DRAM. In two-input NAND gates, where two PMOS transistors are connected in parallel and two NMOS transistors in series, the speed improvement at the inverter level is anticipated to translate to approximately 75% of that in the CMOS inverter. For more complex logic like XOR gates, although precise quantification is challenging due to the increased number of transistors, performance gains are expected qualitatively due to the cumulative transistor-level swing improvements. Additionally, in DRAM devices, the improved subthreshold swing can also enhance data retention and minimize power consumption during operation. These transistor-level improvements consequently form a solid basis for scalable performance enhancements in larger integrated circuits, supporting the broad applicability of this barrier height engineering.

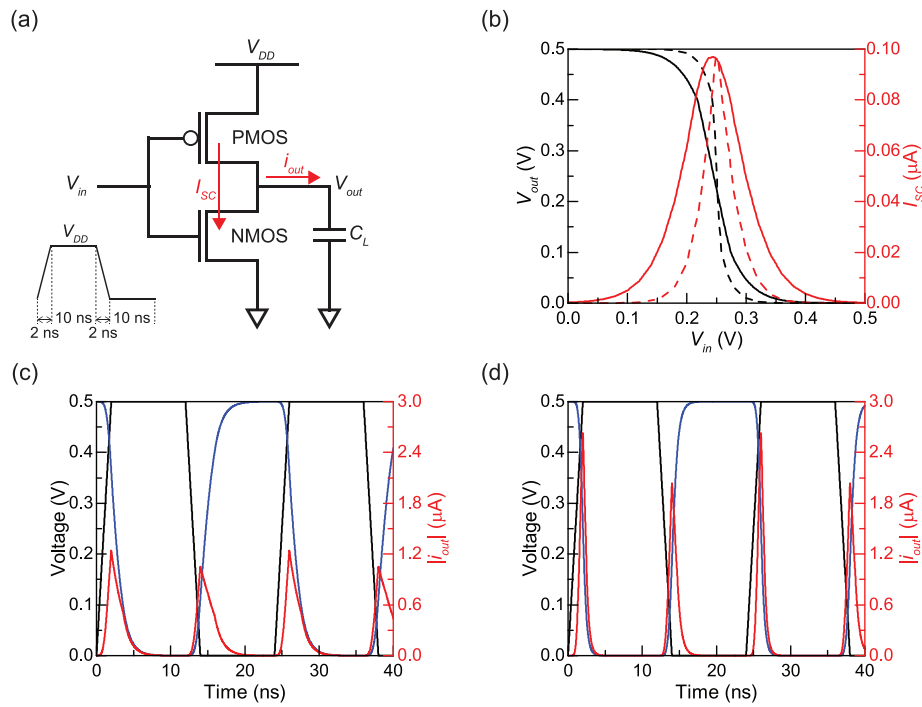


FIGURE 7 | (a) Schematic representation of the CMOS inverter operating at a supply voltage $V_{DD} = 0.5$ V. Both PMOS and NMOS transistors are implemented using bilayer MoS_2 SB-FETs, with a 2 nm HfO_2 gate dielectric and a channel width of 100 nm. (b) Voltage transfer characteristics and short-circuit current curves. Solid lines correspond to the baseline device behavior, while dashed lines indicate the modified characteristics following the application of barrier height reduction. (c) Dynamic response of the CMOS inverter without barrier height reduction. (d) Dynamic response of the CMOS inverter with barrier height reduction applied. In subplots (c) and (d), load capacitance (C_L) was set to 5 fF. Black and blue lines represent input and output voltage, respectively.

2.6 | Practical Feasibility

Bilayer TMDs remain the most practical and promising configuration for exploiting electric-field-induced bandgap narrowing due to their favorable symmetry properties and relative ease of fabrication. For monolayer TMDs, although the bands are distorted under a perpendicular electric field, the lack of inter-layer coupling means that the bandgap remains essentially unchanged [41]. While theoretical studies indicate that tri-layer and four-layer WSe_2 with ideal 2H stacking can also exhibit bandgap narrowing under a vertical electric field [42]—owing to the breaking of z-symmetry in tri-layers and inversion symmetry in four-layers—achieving perfect stacking sequences and uniform layer quality in thicker layered structures presents significant experimental challenges. With increasing layer number, precise control over stacking order and interface quality becomes more difficult, potentially limiting the reproducibility and scalability of bandgap narrowing effects predicted theoretically. In contrast, bilayer TMDs exhibit a robust bandgap narrowing response even when the stacking deviates from the ideal 2H configuration, maintaining the inversion or z-symmetry necessary for the effect [7]. This structural robustness, combined with comparatively less demanding fabrication processes, makes bilayers the most experimentally feasible and scalable system for practical device implementation. Consequently, although multilayer configurations may theoretically expand the tunability range under vertical electric fields, bilayer TMDs offer an optimal balance of tunability, feasibility, and device scalability suitable for emerging nanoelectronic applications

that demand ultra-low power consumption, including wearable technologies [43–45].

This study focuses on simulation-based investigations of bilayer MoS_2 transport properties, yet its significance is reinforced by experimental findings. A previous study demonstrated that bilayer MoS_2 FETs exhibit a subthreshold swing of 164 mV/dec, significantly lower than the 247 mV/dec observed in monolayer counterparts [9], corresponding to an SS improvement of approximately 33.6%. Achieving sub-60 mV/dec SS performance experimentally will require meticulous management of device quality, including interface defects, bubbles, and contact resistance, alongside the formation of high-quality edge contacts to maximize Schottky barrier reduction. Given the sustained and growing interest in bilayer TMD research [46–48], it is anticipated that sub-60 mV/dec SS operation may be experimentally realized in the near future.

3 | Conclusion

Vertical electric field-induced bandgap narrowing in bilayer MoS_2 leads to a reduction of Schottky barrier height at the metal-semiconductor interface, resulting in significant improvement in subthreshold swing (SS) of field-effect transistors. By incorporating DFT-calculated band edge shifts into a custom transport model, the impact on thermionic emission and tunneling currents at the Schottky contact was analyzed. This approach yielded an SS of 44.7 mV/dec in a bilayer MoS_2 SB-FET with a 2 nm HfO_2

gate dielectric, representing a 37.5% reduction. Furthermore, the dynamic response of a CMOS inverter revealed up to a 38% enhancement in switching speed and a reduction in total power consumption by approximately 5%. This mechanism, enabled by the inversion symmetry of bilayer 2D materials and their inherent bandgap response to vertical electric fields, offers a promising pathway toward steep-slope and energy-efficient logic devices.

4 | Methods

4.1 | Density Functional Theory Calculation

The effects of an external electric field on bilayer MoS₂ were investigated using DFT calculations, performed with SIESTA (Spanish Initiative for Electronic Simulations with Thousands of Atoms) [49]. The exchange-correlation energy was described using the Perdew–Burke–Ernzerhof (PBE) functionals [50] under the generalized gradient approximation (GGA). The van der Waals (vdW) correction was implemented by DF2 [51, 52]. The wave functions were expanded using an atomic orbital basis set with a kinetic energy cut-off of 600 Ry, and the first Brillouin zone was sampled by a $24 \times 24 \times 1$ uniform mesh. After the structure relaxation, the lattice constant was found to be 3.148 Å. The atomic coordinates were optimized by minimizing the atomic forces to below 0.001 eV/Å for all cases. Upon geometric relaxation, the interlayer spacing was determined to be 6.20 Å. A vertical cell length of 30 Å was utilized to provide enough vacuum regions above and below the studied structure, mitigating any spurious interactions arising from periodic boundary conditions. To simulate the effects of the external electric field, a constant electric field across the MoS₂ was applied orthogonal to the slab surfaces. Moreover, key physical parameters, including the effective masses, energy bandgap, and layer thickness, were estimated from the DFT results to ensure accurate representation of the material's electronic characteristics in subsequent transport simulations. These parameters are summarized in Table 3.

4.2 | Bilayer MoS₂ FET Transport Model

The transport simulation was performed by implementing the standard set of five Shockley equations: Poisson's equation, electron and hole current density equations, and electron and hole carrier continuity equations, to describe carrier dynamics in bilayer MoS₂. Additionally, the carrier densities were estimated by the Fermi–Dirac distribution without approximation to the Maxwell–Boltzmann in order to account for the wide range of Fermi levels, which is necessary to describe the metal contact area accurately. Specifically, using a discretized mesh, the charge–voltage (Q – V) relationship at each node in a MoS₂ channel is given by:

$$V_{GS} - \psi_i + \frac{q(p_i - n_i + N_D - N_A)}{C_{ox}} - \frac{1}{q}(\varphi_G - \chi_{MoS_2}) + \left(\frac{E_{gap}}{2q} + \frac{k_B T}{q} \ln \frac{N_C}{N_V} \right) + \frac{\epsilon_{MoS_2}}{C_{ox}} \frac{T_{MoS_2}}{\Delta x_i + \Delta x_{i-1}} \left(\frac{\psi_{i-1} - \psi_i}{\Delta x_{i-1}} + \frac{\psi_{i+1} - \psi_i}{\Delta x_i} \right) = 0, \quad (1)$$

TABLE 3 | Parameters and values utilized in transport simulation.

Parameter	Symbol	Value
Effective mass of an electron	m_n^* (K)	$0.63 m_0$
	m_n^* (Σ_{min})	$0.66 m_0$
Effective mass of a hole	m_p^* (Γ)	$1.07 m_0$
	m_p^* (K)	$0.68 m_0$
Energy bandgap	E_{gap}	1.4 eV
Work function of gate metal (Ti)	φ_G	4.3 eV
Relative permittivity [59] of bilayer MoS ₂	ϵ_{r,MoS_2}	4
Thickness of the bilayer MoS ₂	T_{MoS_2}	9.4 Å
Relative permittivity of HfO ₂	ϵ_{r,HfO_2}	25
Electron mobility [60] of bilayer MoS ₂	μ_n	100 cm ² /V·s
Hole mobility of bilayer MoS ₂	μ_p	63 cm ² /V·s

which incorporates the influence of the gate bias via Poisson's equation. This approach is similar to that used for graphene FET [53], another widely studied 2D crystal, except that a bandgap is present in bilayer MoS₂. Here, V_{GS} represents the gate bias, while ψ_i denotes the electrostatic potential of i th node. The variables n and p represent the electron and hole densities, respectively. N_D and N_A are donor and acceptor densities, respectively. The capacitance of the gate oxide, C_{ox} , is defined as ϵ_{ox}/t_{ox} , where ϵ_{ox} and t_{ox} are the permittivity and the thickness of the gate oxide. φ_G is the work function of the gate metal, while χ_{MoS_2} and E_{gap} are the electron affinity and the energy bandgap, respectively. N_C and N_V are the effective density of states of the conduction and valence bands, respectively. The corresponding parameters are summarized in Table 3.

4.3 | Metal-Semiconductor Schottky Contact Transport

There are two distinct transport mechanisms at metal–MoS₂ contacts. One is the thermionic emission current (J_{TE}) and the other quantum-mechanical tunneling current of an electron (J_{TN}). As for J_{TE} , the thermionic emission-diffusion model was adopted [54], which reads as follows for electrons:

$$J_{TE,n} = qv_n(n_s - n_0) \quad (2)$$

and for holes,

$$J_{TE,p} = qv_p(p_s - p_0) \quad (3)$$

Here, $v_{n,p} = (k_B T / 2m_{DOS}^* \pi)^{1/2}$, which are the surface recombination velocities for electrons and holes. The equilibrium electron density (n_0) and hole density (p_0) were calculated using the relations $n_0 = N_C \cdot \mathcal{F}_0(-\frac{q\phi_{b,n}}{kT})$ and $p_0 = N_V \cdot \mathcal{F}_0(-\frac{q\phi_{b,p}}{kT})$, where $\mathcal{F}_0$ denotes the Fermi–Dirac integral of order zero. For Schottky

barrier heights of $q\phi_{b,n} = q\phi_{b,p} = 0.7$ eV, the estimated equilibrium carrier densities were $n_0 = 0.24 \times 10^{12} \text{ cm}^{-2}$ and $p_0 = 0.20 \times 10^{12} \text{ cm}^{-2}$ without a vertical electric field. However, the equilibrium carrier densities can vary with the reduction of the Schottky barrier height. The actual surface densities (n_s and p_s) under specific applied bias conditions were determined from the self-consistent solution of the transport equations.

As for J_{TN} , the universal Schottky tunneling (UST) model was employed [55, 56]. In this model, carrier tunneling from the metal to the semiconductor is treated as local carrier generation in the semiconductor, while tunneling from the semiconductor to the metal is considered as local carrier recombination. Thus, when $G_{TN,u}(x_i)$ is the local tunneling generation rate, and $\theta_u(x_i)$ is the local tunneling probability. Using Wentzel-Kramers-Brillouin (WKB) approximation, these quantities are expressed as:

$$G_{TN,u}(x_i) = \frac{A_{2D,u}^* T^{1/2}}{k_B} \vec{\epsilon}_i \theta_u(x_i) Q_u(x_i) \quad (3)$$

where the tunneling probabilities for electrons and holes are given by:

$$\theta_n(x_i) = \exp \left(-\frac{4\pi \sqrt{2m_{n,DOS}^*} x_i}{h} \int_0^{x_i} \sqrt{E_C(x') - E_C(x_i)} dx' \right) \quad (4)$$

$$\theta_p(x_i) = \exp \left(-\frac{4\pi \sqrt{2m_{p,DOS}^*} x_i}{h} \int_0^{x_i} \sqrt{E_V(x_i) - E_V(x')} dx' \right) \quad (5)$$

and the local supply function $Q(x_i)$ is defined as:

$$Q_n(x_i) = \left(\mathcal{F}_0 \left(\frac{E_{FM} - E_{C,i}}{k_B T} \right) - \mathcal{F}_0 \left(\frac{F_{N,i} - E_{C,i}}{k_B T} \right) \right) \quad (6)$$

$$Q_p(x_i) = \left(\mathcal{F}_0 \left(\frac{E_{FM} - E_{V,i}}{k_B T} \right) - \mathcal{F}_0 \left(\frac{F_{p,i} - E_{V,i}}{k_B T} \right) \right) \quad (7)$$

Here, subscript u denotes the carrier type, with n for electron and p for hole [55–57]. The coefficient $A_{2D,u}^* = q(8\pi m_{u,DOS}^* k_B^3)^{1/2} / h^2$ is Richardson constant for 2-D materials [21, 58]. $\vec{\epsilon}_i$ is the local electric field intensity. E_C and E_V are the conduction and valence band edges of the semiconductor, respectively. E_{FM} is the Fermi level of the contact metal, while $F_{N,i}$ and $F_{p,i}$ are the local quasi-Fermi levels for electrons and holes in the semiconductor, respectively. The tunneling current density at the Schottky contact is then obtained by integrating the local tunneling generation rate:

$$J_{TN,u} = q \int_0^{W_{TN,u}} G_{TN,u}(x') dx' \quad (8)$$

where W_{TN} denotes the width of the tunneling layer.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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